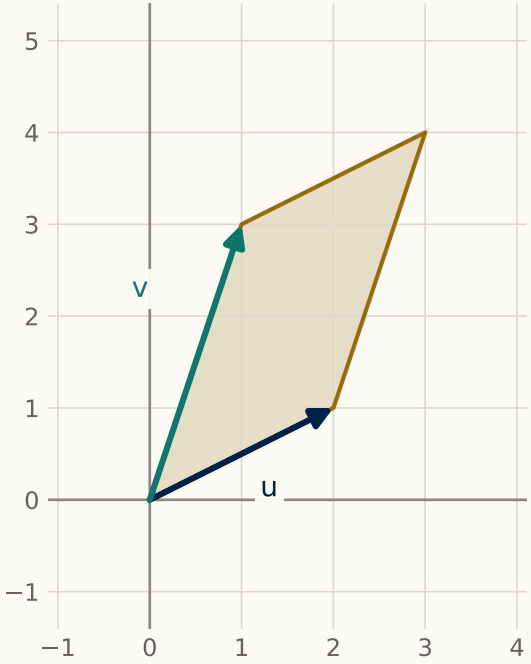
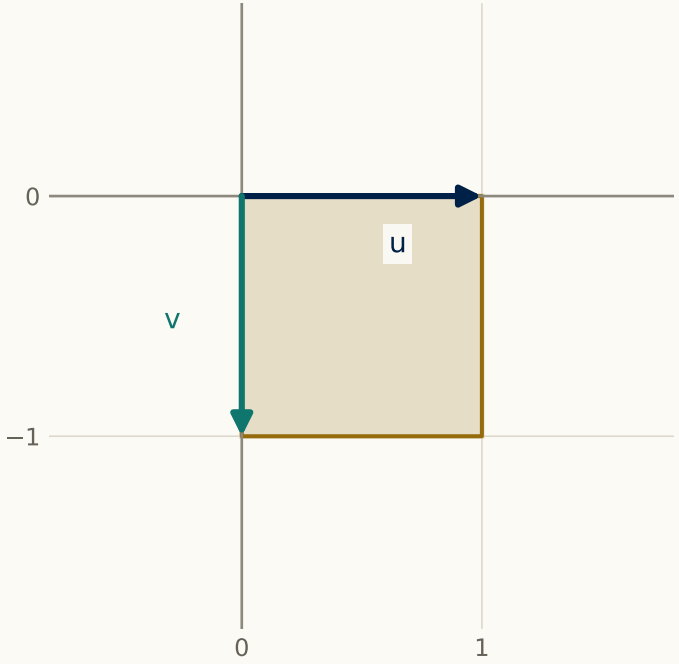
Signed area remembers orientation

det = 2·3 − 1·1 = 5



reflection: det = −1